



Mansoura University
Faculty of Computers and Information
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MODELLING & SIMULATION

SW,BI,AI

Prepared by:

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COURSE SYLLABUS

- **Chapter 1: Introduction.**
- **Chapter 2: Probability as Using in Simulation.**
- **Chapter 3: Queueing Simulation.**
- **Chapter 4: Inventory Simulation.**
- **Chapter 5: Random-Number Generation.**
- **Chapter 6: Input Modeling.**
- **Chapter 7: Output Analysis.**

COURSE SYLLABUS

- **Chapter 1: Introduction.**
- **Chapter 2: Probability as Using in Simulation.**
- **Chapter 3: Queueing Simulation (Part 2)**
- **Chapter 4: Inventory Simulation.**
- **Chapter 5: Random-Number Generation.**
- **Chapter 6: Input Modeling.**
- **Chapter 7: Output Analysis.**

CHAPTER CONTENT: QUEUEING SIMULATION

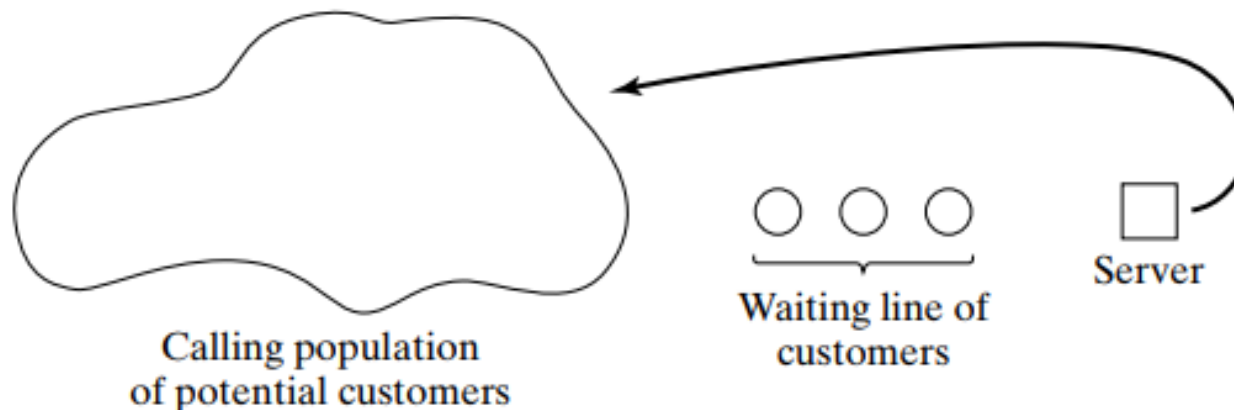
- Introduction.
- Description of Queuing System.
- Simulating a Single-Server Queue.
 - Example.
- Simulating a Queue with Multi-Servers.
 - Example.

CHAPTER CONTENT: QUEUEING SIMULATION

- Introduction.
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- Simulating a Queue with Multi-Servers.
 - Example.

INTRODUCTION

- **Queueing Model:**
- Queueing models are among the most **widely used applications of simulation** in practice.
- Simulation is often used in the **analysis** of queueing models.
- In a simple but typical queueing model, customers arrive from time to time and join a queue (waiting line), are eventually served, and finally leave the system.



DESCRIPTION OF QUEUING SYSTEM

- The key elements of a queueing system are the **customers** and **servers**.

Element	Definition	Examples
Customers	Anything requiring service	People, machines, trucks, patients, airplanes, e-mail
Servers	Resources providing service	Receptionists, repair personnel, medical staff, runways

TYPICAL MEASURES OF PERFORMANCE

- When analyzing queueing systems, we typically focus on several key performance measures:

Performance Measure	Description
Server Utilization	Percentage of time a server is busy
Queue Length	Number of customers waiting in line
Customer Delays	Time customers spend waiting before service
System Time	Total time customers spend in the system (waiting + service)
Probability of Waiting	Likelihood that an arriving customer must wait

DESCRIPTION OF QUEUING SYSTEM

■ Queueing System:

- 1) The Calling Population.
- 2) System Capacity.
- 3) The Arrival Process.
- 4) Queue Behavior and Queue Discipline.
- 5) Service Times and the Service Mechanism.

DESCRIPTION OF QUEUING SYSTEM

- A notational system for parallel server systems: A/B/c/N/K.
- These letters represent the following system characteristics:

Symbol	Meaning	Common Examples
A	Arrival time distribution	M (Exponential), D (Deterministic), G (General)
B	Service time distribution	M (Exponential), D (Deterministic), G (General)
c	Number of parallel servers	1, 2, 3, ...
N	System capacity (optional)	If omitted, assumed infinite
K	Size of calling population (optional)	If omitted, assumed infinite

KEY CALCULATIONS

- **For each customer, we compute:**

1. **Arrival Time:** Arrival Time of previous customer + Time Between Arrivals
2. **Time Service Begins:** $\max(\text{Arrival Time}, \text{Time Service Ends of previous customer})$
3. **Time Service Ends:** Time Service Begins + Service Time
4. **Waiting Time in Queue:** Time Service Begins - Arrival Time
5. **Time Spent in System:** Time Service Ends - Arrival Time (or Waiting Time + Service Time)
6. **Server Idle Time:** Time Service Begins - Time Service Ends of previous customer (if positive)

Simulation System

Performance Measure

Customer	Interarrival Time [Minutes]	Arrival Time [Clock]	Service Time [Minutes]	Time Service Begins [Clock]	Time Service Ends [Clock]	Waiting Time in Queue [Minutes]	Time Customer in System [Minutes]	Idle Time of Server [Minutes]
1	—	0	4	0	4	0	4	0
2	8	8	1	8	9	0	1	4
3	6	14	4	14	18	0	4	5
4	1	15	3	18	21	3	6	0
5	8	23	2	23	25	0	2	2
6	3	26	4	26	30	0	4	1
7	8	34	5	34	39	0	5	4
8	7	41	4	41	45	0	4	2
9	2	43	5	45	50	2	7	0
10	3	46	3	50	53	4	7	0
11	1	47	3	53	56	6	9	0
12	1	48	5	56	61	8	13	0
13	5	53	4	61	65	8	12	0
14	6	59	1	65	66	6	7	0
15	3	62	5	66	71	4	9	0
16	8	70	4	71	75	1	5	0
17	1	71	3	75	78	4	7	0
18	2	73	3	78	81	5	8	0
19	4	77	2	81	83	4	6	0
20	5	82	3	83	86	1	4	0
Total	82		68			56	124	18

SOME OF THE FINDINGS FROM THE SIMULATION ARE AS FOLLOWS

$$\text{average waiting time (minutes)} = \frac{\text{total time customers wait in queue (minutes)}}{\text{total numbers of customers}}$$

$$= \frac{56}{20} = 2.8 \text{ minutes}$$

SOME OF THE FINDINGS FROM THE SIMULATION ARE AS FOLLOWS

$$\begin{aligned}\text{probability (wait)} &= \frac{\text{number of customers who wait}}{\text{total number of customers}} \\ &= \frac{13}{20} = 0.65\end{aligned}$$

SOME OF THE FINDINGS FROM THE SIMULATION ARE AS FOLLOWS

$$\begin{array}{l} \text{probability of idle} \\ \text{server} \end{array} = \frac{\text{total idle time of server (minutes)}}{\text{total run time of simulation (minutes)}}$$

$$= \frac{18}{86} = 0.21$$

SIM.A SINGLE-SERVER QUEUE

- Our objective is to analyze the system by simulating the arrival and service of 100 customers and to compute a variety of typical measures of performance for queueing models.
 - In actuality, 100 customers may be too small a sample size to draw **reliable conclusions**. Depending on our objectives, the accuracy of the results may be enhanced by increasing the sample size (number of customers), or by running multiple trials (or replications).

SIM.A SINGLE-SERVER QUEUE

- Our objective is to analyze the system by simulating the arrival and service of 100 customers and to compute a variety of typical measures of performance for queueing models.
 - **A second issue is that of initial conditions.** A simulation of a grocery store that starts with an empty system may or may not be realistic unless the intention is to model the system from startup or to model until steady-state operation is reached. Here, to keep calculations simple, the starting conditions are an empty grocery, and any concerns are overlooked.

SIMULATION TABLE FOR QUEUEING PROBLEM (100 CUSTOMERS)

Simulation System

Performance Measure

Customer	Interarrival Time [Minutes]	Arrival Time [Clock]	Service Time [Minutes]	Time Service Begins [Clock]	Time Service Ends [Clock]	Waiting Time in Queue [Minutes]	Time Customer in System [Minutes]	Idle Time of Server [Minutes]
1	-	0	4	0	4	0	4	0
2	1	1	2	4	6	3	5	0
3	1	2	5	6	11	4	9	0
4	6	8	4	11	15	3	7	0
5	3	11	1	15	16	4	5	0
6	7	18	5	18	23	0	5	2
...								
100	5	415	2	416	418	1	3	0
Total	415		317			174	491	101

SOME OF THE FINDINGS FROM THE SIMULATION ARE AS FOLLOWS

Average waiting time

$$\bar{w} = \frac{\sum \text{Waiting time in queue}}{\text{Number of customers}} = \frac{174}{100} = 1.74 \text{ min}$$

Probability that a customer has to wait

$$p(\text{wait}) = \frac{\text{Number of customer who wait}}{\text{Number of customers}} = \frac{46}{100} = 0.46$$

Proportion of server idle time

$$p(\text{idle server}) = \frac{\sum \text{Idle time of server}}{\text{Simulation run time}} = \frac{101}{418} = 0.24$$

Average service time

$$\bar{s} = \frac{\sum \text{Service time}}{\text{Number of customers}} = \frac{317}{100} = 3.17 \text{ min}$$

$$E(s) = \sum_{s=0}^{\infty} s \cdot p(s) = 0.1 \cdot 10 + 0.2 \cdot 20 + \dots + 0.05 \cdot 6 = 3.2 \text{ min}$$

SOME OF THE FINDINGS FROM THE SIMULATION ARE AS FOLLOWS

Average time between arrivals

$$\bar{\lambda} = \frac{\sum \text{Times between arrivals}}{\text{Number of arrivals} - 1} = \frac{415}{99} = 4.19 \text{ min}$$

$$E(\lambda) = \frac{a+b}{2} = \frac{1+8}{2} = 4.5 \text{ min}$$

Average waiting time of those who wait

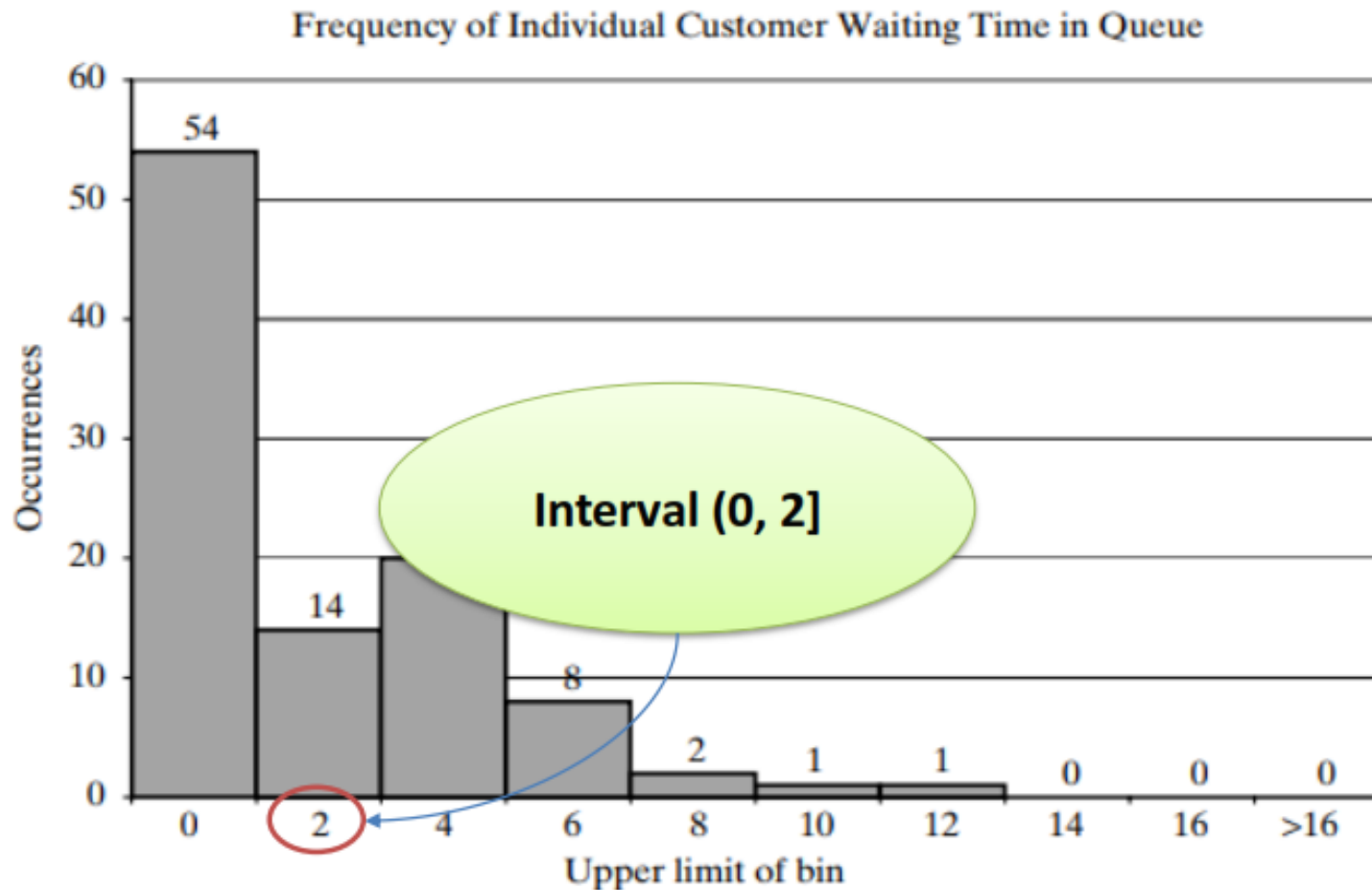
$$\bar{w}_{\text{waited}} = \frac{\sum \text{Waiting time in queue}}{\text{Number of customers that wait}} = \frac{174}{54} = 3.22 \text{ min}$$

Average time a customer spends in system

$$\bar{t} = \frac{\sum \text{Time customers spend in system}}{\text{Number of customers}} = \frac{491}{100} = 4.91 \text{ min}$$

$$\bar{t} = \bar{w} + \bar{s} = 1.74 + 3.17 = 4.91 \text{ min}$$

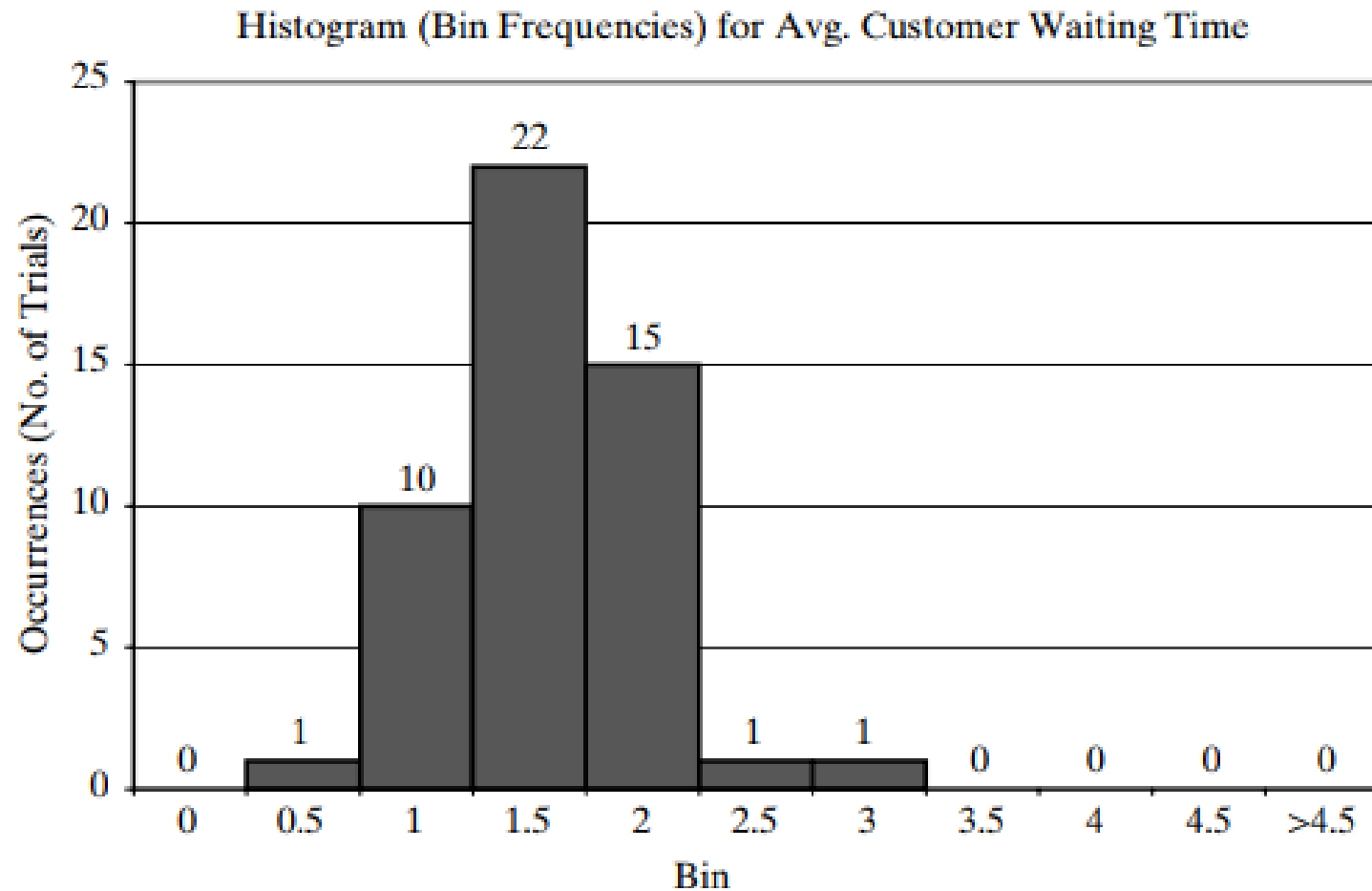
SOME OF THE FINDINGS FROM THE SIMULATION ARE AS FOLLOWS



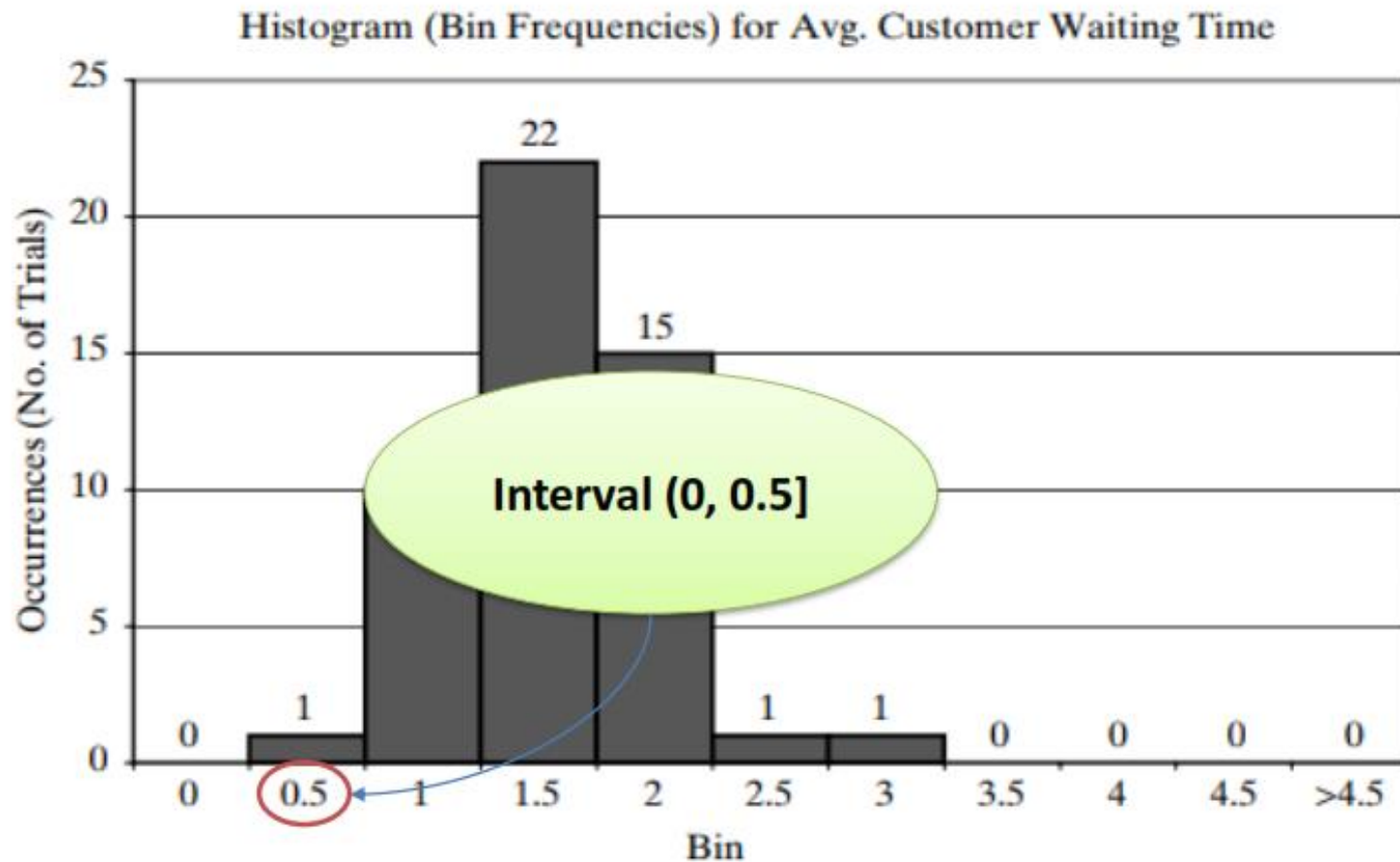
SOME OF THE FINDINGS FROM THE SIMULATION ARE AS FOLLOWS

- We are now interested in running an experiment to address the question of how the average time in queue for the first 100 customers varies from day to day.
- Running once with 100 customers corresponds to one day. Running it 50 times corresponds to 50 days, where each trial represents one day.
- The overall average waiting time over 50 trials was 1.32 minutes.

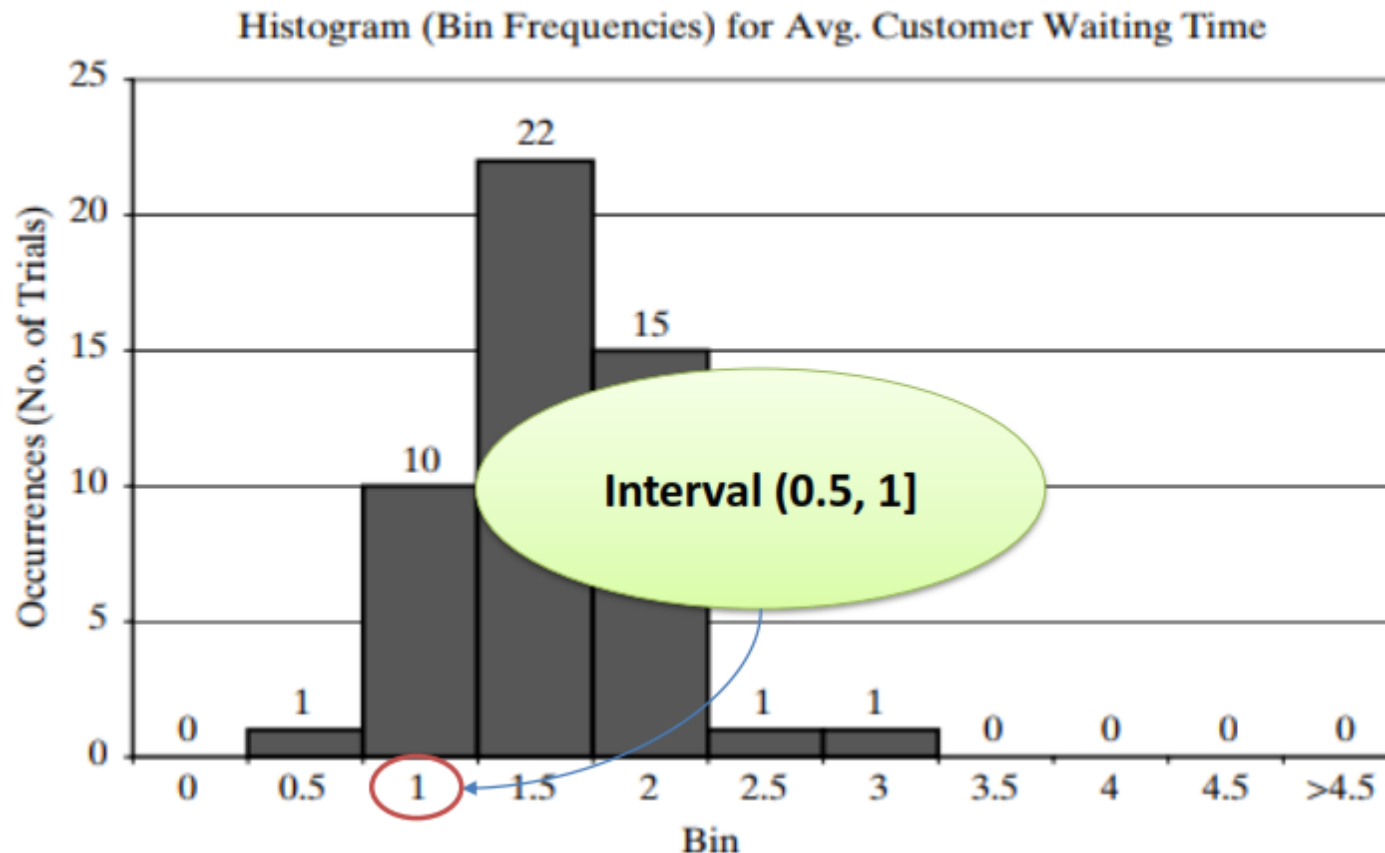
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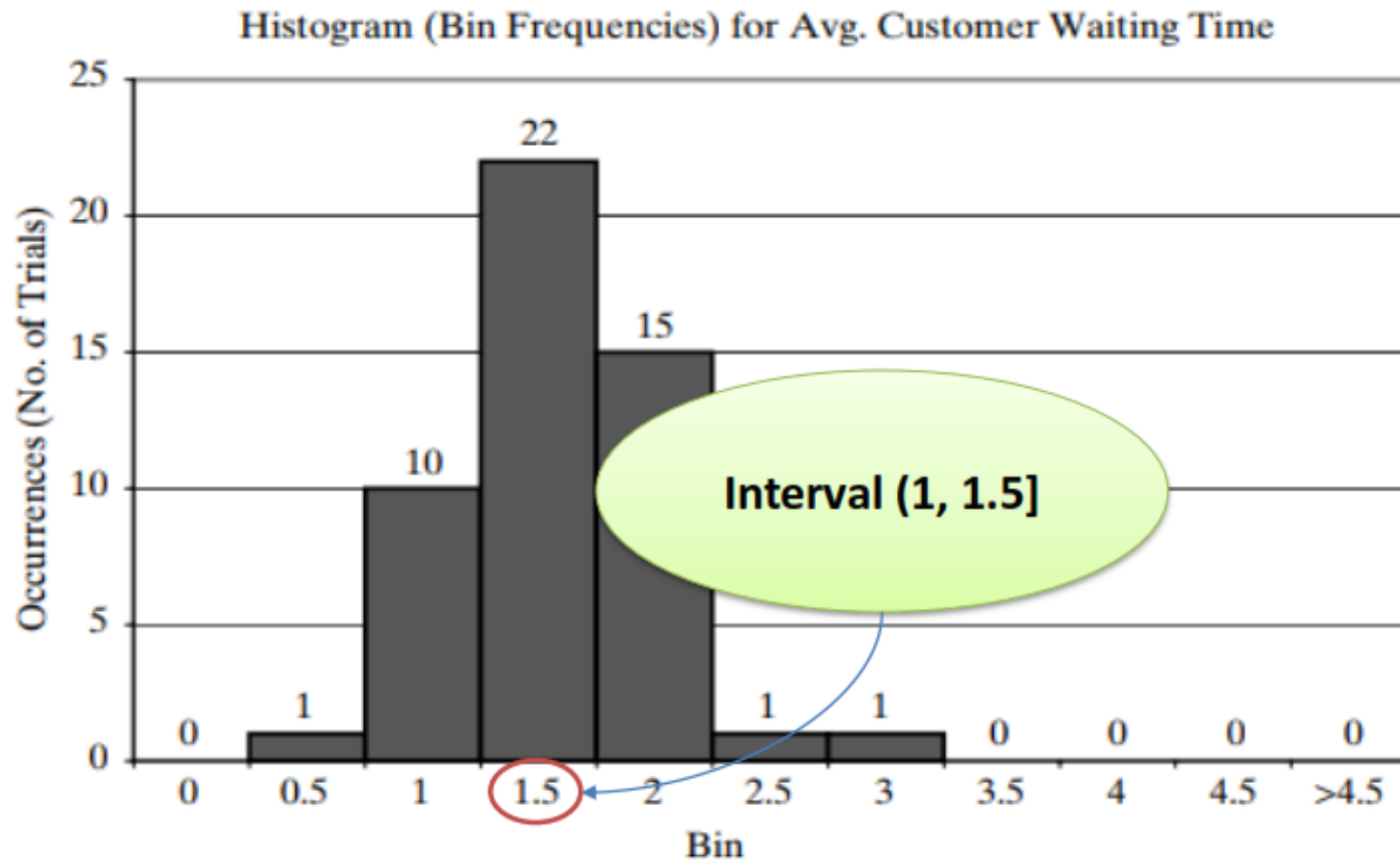
SOME OF THE FINDINGS FROM THE SIMULATION ARE AS FOLLOWS



SOME OF THE FINDINGS FROM THE SIMULATION ARE AS FOLLOWS



SOME OF THE FINDINGS FROM THE SIMULATION ARE AS FOLLOWS



Quiz

Time between Arrivals (Minutes)	Probability
1	0.5
3	0.3
6	0.2

Service Time (Minutes)	Probability
1	0.35
3	0.55
5	0.10

- **Calculate** for each customer the arrival time, service starting and ending times, waiting time in queue, time spent in the system, and idle time in the server. Show the results in the simulation table.

Time-Between-Arrivals Determination		Service Times Generated	
Customer	Random Digits	Customer	Random Digits
1	–	1	5
2	0	2	45
3	5	3	66
4	3	4	32
5	7	5	80

- **Compute** average waiting time, probability (wait), probability of idle server, average waiting time for those who wait, average service time, average time between arrivals

Time between Arrivals (Minutes)	Probability	Cumulative Probability	Random- Digit Assignment
1	0.5		
3	0.3		
6	0.2		

Time between Arrivals (Minutes)	Probability	Cumulative Probability	Random- Digit Assignment
1	0.5		
3	0.3		
6	0.2		

Time between Arrivals (Minutes)	Probability	Cumulative Probability	Random- Digit Assignment
1	0.5	0.5	1-5
3	0.3	0.8	6-8
6	0.2	1.0	9-0

Service Time (Minutes)	Probability	Cumulative Probability	Random-Digit Assignment
1	0.35		
3	0.55		
5	0.10		

Service Time (Minutes)	Probability	Cumulative Probability	Random-Digit Assignment
1	0.35		
3	0.55		
5	0.10		

Service Time (Minutes)	Probability	Cumulative Probability	Random-Digit Assignment
1	0.35	0.35	01-35
3	0.55	0.90	36-90
5	0.10	1.00	91-00

Simulation Table

Customer	Time Since Last Arrival (Minutes)	Arrival Time	Service Time (Minutes)	Time Service Begins	Time Service Ends
1					
2					
3					
4					
5					

Time-Between-Arrivals Determination		Service Times Generated	
Customer	Random Digits	Customer	Random Digits
1	—	1	5
2	0	2	45
3	5	3	66
4	3	4	32
5	7	5	80

Simulation Table

Customer	Time Since Last Arrival (Minutes)	Arrival Time	Service Time (Minutes)	Time Service Begins	Time Service Ends
1	–	0	1	0	1
2	6	6	3	6	9
3	1	7	3	9	12
4	1	8	1	12	13
5	3	11	3	13	16

Time-Between-Arrivals Determination		Service Times Generated	
Customer	Random Digits	Customer	Random Digits
1	–	1	5
2	0	2	45
3	5	3	66
4	3	4	32
5	7	5	80

Customer	Time Since Last Arrival (Minutes)	Arrival Time	Service Time (Minutes)	Time Service Begins	Time Service Ends	Time Customer Waits in Queue (Minutes)	Time Customer Spends in System (Minutes)	Idle Time of Server (Minutes)
1	–	0	1	0	1			
2	6	6	3	6	9			
3	1	7	3	9	12			
4	1	8	1	12	13			
5	3	11	3	13	16			

Customer	Time Since Last Arrival (Minutes)	Arrival Time	Service Time (Minutes)	Time Service Begins	Time Service Ends	Time Customer Waits in Queue (Minutes)	Time Customer Spends in System (Minutes)	Idle Time of Server (Minutes)
1	–	0	1	0	1	0	1	0
2	6	6	3	6	9	0	3	5
3	1	7	3	9	12	2	5	0
4	1	8	1	12	13	4	5	0
5	3	11	3	13	16	2	5	0

Customer	Time Since Last Arrival (Minutes)	Arrival Time	Service Time (Minutes)	Time Service Begins	Time Service Ends	Time Customer Waits in Queue (Minutes)	Time Customer Spends in System (Minutes)	Idle Time of Server (Minutes)
1	–	0	1	0	1	0	1	0
2	6	6	3	6	9	0	3	5
3	1	7	3	9	12	2	5	0
4	1	8	1	12	13	4	5	0
5	3	11	3	13	16	2	5	0
Total	11		11			8	19	5


$$\text{average waiting time (minutes)} = \frac{\text{total time customers wait in queue (minutes)}}{\text{total numbers of customers}}$$

$$\text{probability (wait)} = \frac{\text{number of customers who wait}}{\text{total number of customers}}$$

$$\text{probability of idle server} = \frac{\text{total idle time of server (minutes)}}{\text{total run time of simulation (minutes)}}$$

$$\text{Average waiting time of those who wait (minutes)} = \frac{\text{total time customers wait in queue (minutes)}}{\text{total number of customers who wait}}$$

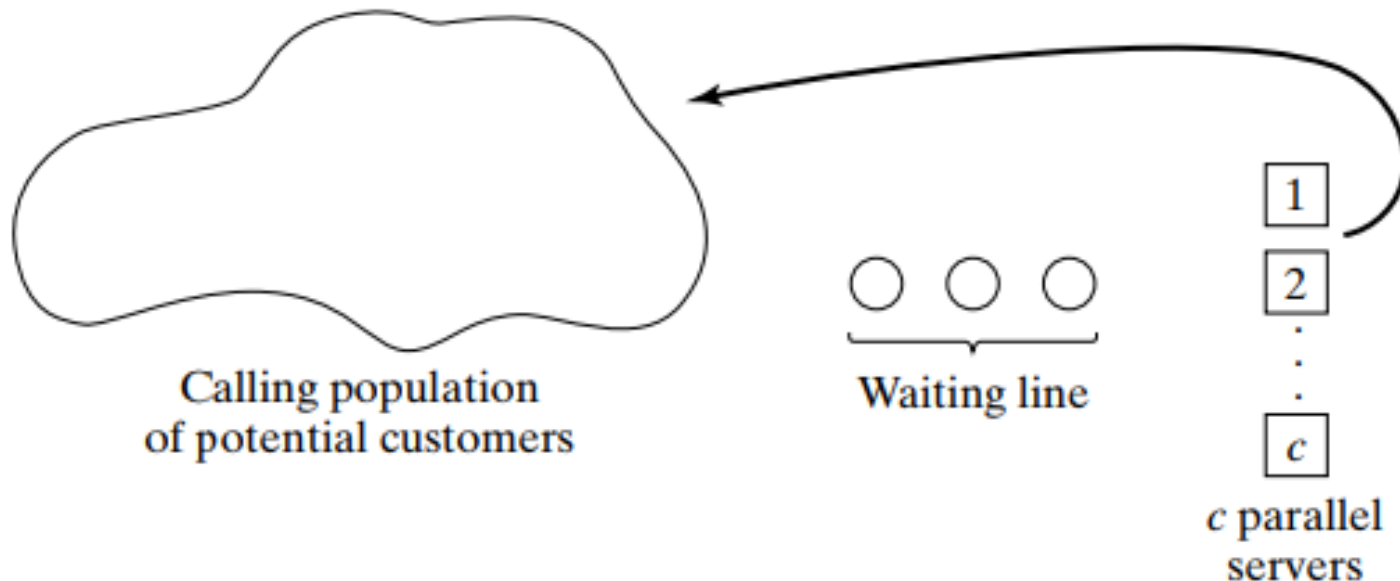
$$\text{average service time (minutes)} = \frac{\text{total service time (minutes)}}{\text{total number of customers}}$$

$$\text{average time between arrivals (minutes)} = \frac{\text{sum of all times between arrivals (minutes)}}{\text{number of arrivals} - 1}$$

SIM.A MULTI-SERVER QUEUE

■ Queueing Model:

- Suppose that there are c channels operating in parallel. Arrivals will join a single queue and enter the first available service channel.



EXAMPLE: CALL CENTER:



EXAMPLE: CALL CENTER:

- **Example: Call Center:**
- Consider a Call Center where technical personnel take calls and provide service.
- Two technical support people (2 server) exists:
 - *Able* more experienced, provides service faster,
 - *Baker* newbie, provides service slower.
- **Rule:**
 - Able gets call if both people are idle.

EXAMPLE: CALL CENTER:

- Interarrival distribution of calls for technical support

<i>Time between Arrivals (Minutes)</i>	<i>Probability</i>	<i>Cumulative Probability</i>	<i>Random-Digit Assignment</i>
1	0.25	0.25	01–25
2	0.40	0.65	26–65
3	0.20	0.85	66–85
4	0.15	1.00	86–00

EXAMPLE: CALL CENTER:

■ Service Distribution of Able

<i>Service Time (Minutes)</i>	<i>Probability</i>	<i>Cumulative Probability</i>	<i>Random-Digit Assignment</i>
2	0.30	0.30	01–30
3	0.28	0.58	31–58
4	0.25	0.83	59–83
5	0.17	1.00	84–00

■ Service Distribution of Baker

<i>Service Time (Minutes)</i>	<i>Probability</i>	<i>Cumulative Probability</i>	<i>Random-Digit Assignment</i>
3	0.35	0.35	01–35
4	0.25	0.60	36–60
5	0.20	0.80	61–80
6	0.20	1.00	81–00

EXAMPLE: CALL CENTER:

- **The simulation:**
- The problem is to find how well the current arrangement is working. To estimate the system measures of performance, a simulation of **1 hour of operation is made**. A longer simulation would yield more reliable results, but for purposes of illustration a 1-hour period has been selected.

EXAMPLE: CALL CENTER:

Simulation Table:

A	B	C	D	E	F	G	H	I	J	K	L
Customer	Random Digits	Time between	Clock Time	Random Digits	Able		Baker				
No.	for Arrival	Arrivals	of Arrival	for Service	Time Service Begins	Service Time	Time Service Ends	Time Service Begins	Service Time	Time Service Ends	Time in Queue
1	—	—	0	95	0	5	5				0
2	26	2	2	21				2	3	5	0
3	98	4	6	51	6	3	9				0
4	90	4	10	92	10	5	15				0
5	26	2	12	89				12	6	18	0

Service Time (Minutes)	Probability	Cumulative Probability	Random-Digit Assignment
3	0.35	0.35	01–35
4	0.25	0.60	36–60
5	0.20	0.80	61–80
6	0.20	1.00	81–00

Service Time (Minutes)	Probability	Cumulative Probability	Random-Digit Assignment
2	0.30	0.30	01–30
3	0.28	0.58	31–58
4	0.25	0.83	59–83
5	0.17	1.00	84–00

EXAMPLE: CALL CENTER:

Simulation Table:

A	B	C	D	E	F	G	H	I	J	K	L
Customer	Random Digits	Time between	Clock Time	Random Digits	Able		Baker				
No.	for Arrival	Arrivals	of Arrival	for Service	Time Service Begins	Service Time	Time Service Ends	Time Service Begins	Service Time	Time Service Ends	Time in Queue
1	—	—	0	95	0	5	5				0
2	26	2	2	21				2	3	5	0
3	98	4	6	51	6	3	9				0
4	90	4	10	92	10	5	15				0
5	26	2	12	89				12	6	18	0

Baker

Service Time (Minutes)	Probability	Cumulative Probability	Random-Digit Assignment
3	0.35	0.35	01–35
4	0.25	0.60	36–60
5	0.20	0.80	61–80
6	0.20	1.00	81–00

Able

Service Time (Minutes)	Probability	Cumulative Probability	Random-Digit Assignment
2	0.30	0.30	01–30
3	0.28	0.58	31–58
4	0.25	0.83	59–83
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EXAMPLE: CALL CENTER:

Simulation Table:

A	B	C	D	E	F	G	H	I	J	K	L
Customer	Random Digits	Time between	Clock Time	Random Digits	Able		Baker				
No.	for Arrival	Arrivals	of Arrival	for Service	Time Service Begins	Service Time	Time Service Ends	Time Service Begins	Service Time	Time Service Ends	Time in Queue
1	—	—	0	95	0	5	5				0
2	26	2	2	21				2	3	5	0
3	98	4	6	51	6	3	9				0
4	90	4	10	92	10	5	15				0
5	26	2	12	89				12	6	18	0

Baker

Service Time (Minutes)	Probability	Cumulative Probability	Random-Digit Assignment
3	0.35	0.35	01–35
4	0.25	0.60	36–60
5	0.20	0.80	61–80
6	0.20	1.00	81–00

Able

Service Time (Minutes)	Probability	Cumulative Probability	Random-Digit Assignment
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3	0.28	0.58	31–58
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EXAMPLE: CALL CENTER:

Simulation Table:

A	B	C	D	E	F	G	H	I	J	K	L
Customer No.	Random Digits for Arrival	Time between Arrivals	Clock Time of Arrival	Random Digits for Service	Time Service Begins	Service Time	Time Service Ends	Time Service Begins	Service Time	Time Service Ends	Time in Queue
1	—	—	0	95	0	5	5				0
2	26	2	2	21				2	3	5	0
3	98	4	6	51	6	3	9				0
4	90	4	10	92	10	5	15				0
5	26	2	12	89				12	6	18	0

Baker

Service Time (Minutes)	Probability	Cumulative Probability	Random-Digit Assignment
3	0.35	0.35	01–35
4	0.25	0.60	36–60
5	0.20	0.80	61–80
6	0.20	1.00	81–00

Able

Service Time (Minutes)	Probability	Cumulative Probability	Random-Digit Assignment
2	0.35	0.35	01–30
3	0.25	0.60	31–58
4	0.20	0.80	59–83
5	0.20	1.00	84–00

A	B	C	D	E	F	G	H	I	J	K	L
Customer No.	Random Digits for Arrival	Time between Arrivals	Clock Time of Arrival	Random Digits for Service	Able		Time Service Ends	Baker		Time Service Ends	Time in Queue
					Time Service Begins	Service Time		Time Service Begins	Service Time		
1	—	—	0	95	0	5	5				0
2	26	2	2	21				2	3	5	0
3	98	4	6	51	6	3	9				0
4	90	4	10	92	10	5	15				0
5	26	2	12	89				12	6	18	0
6	42	2	14	38	15	3	18				1
7	74	3	17	13	18	2	20				1
8	80	3	20	61	20	4	24				0
9	68	3	23	50				23	4	27	0
10	22	1	24	49	24	3	27				0
11	48	2	26	39	27	3	30				1
12	34	2	28	53				28	4	32	0
13	45	2	30	88	30	5	35				0
14	24	1	31	01				32	3	35	1
15	34	2	33	81	35	4	39				2
16	63	2	35	53				35	4	39	0
17	38	2	37	81	39	4	43				2
18	80	3	40	64				40	5	45	0
19	42	2	42	01	43	2	45				1
20	56	2	44	67	45	4	49				1
21	89	4	48	01				48	3	51	0
22	18	1	49	47	49	3	52				0
23	51	2	51	75				51	5	56	0
24	71	3	54	57	54	3	57				0
25	16	1	55	87				56	6	62	1
26	92	4	59	47	59	3	62				0
						56		43			11

Simulation Table:

EXAMPLE: CALL CENTER:

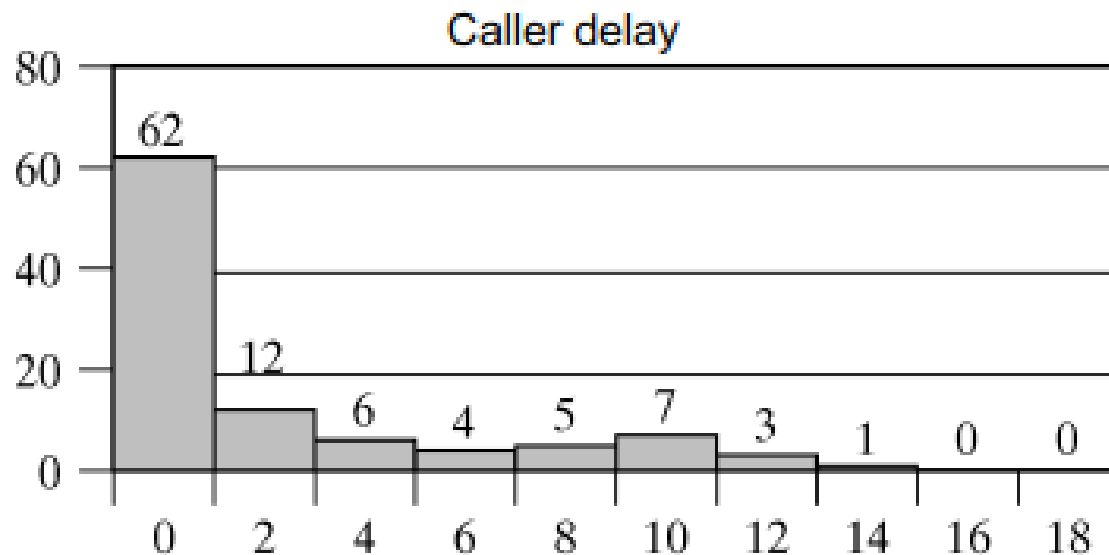
- **The analysis of simulation table results in the following:**
 - 1) Over the 62-minute period Able was busy 90% of the time.
 - 2) Baker was busy only 69% of the time. The seniority rule keeps Baker less busy (and gives Able more tips).
 - 3) Nine of the 26 arrivals (about 35%) had to wait. The average waiting time for all customers was only about 0.42 minute (25 seconds), which is very small.

EXAMPLE: CALL CENTER:

- **The analysis of simulation table results in the following:**
 - 4) Those nine who did have to wait only waited an average of 1.22 minutes, which is quite low.
 - 5) In summary, this system seems well balanced. One server cannot handle all the calls, and three servers would probably be too many. Adding an additional server would surely reduce the waiting time to nearly zero. However, the cost of waiting would have to be quite high to justify an additional server.

SIMULATION RUN FOR 100 CALLS:

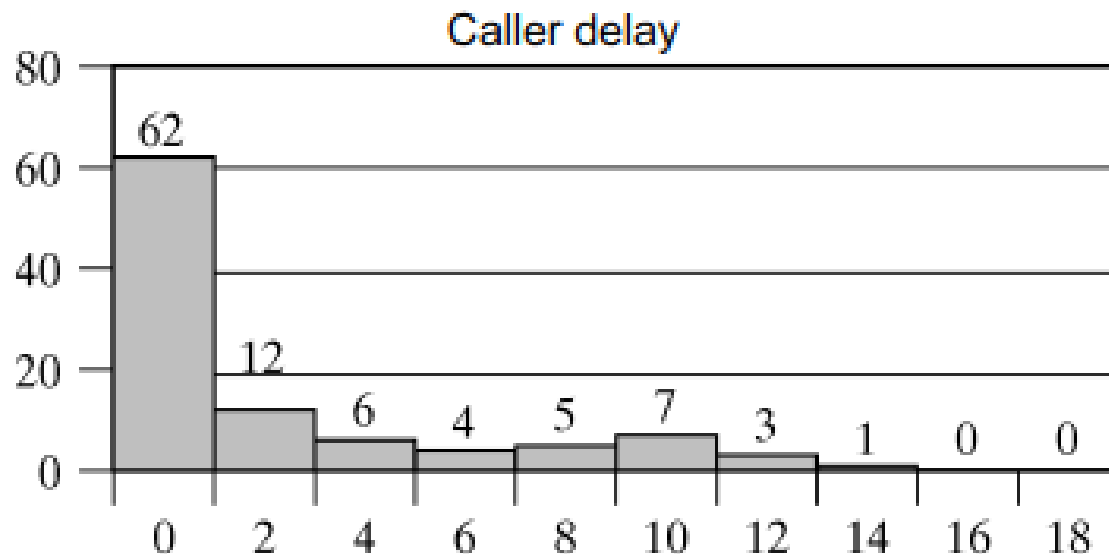
- 62% of callers had no delay.
- 12% of callers had a delay up to 2 minutes.



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**For example:
Average waiting time
= 2 minutes**

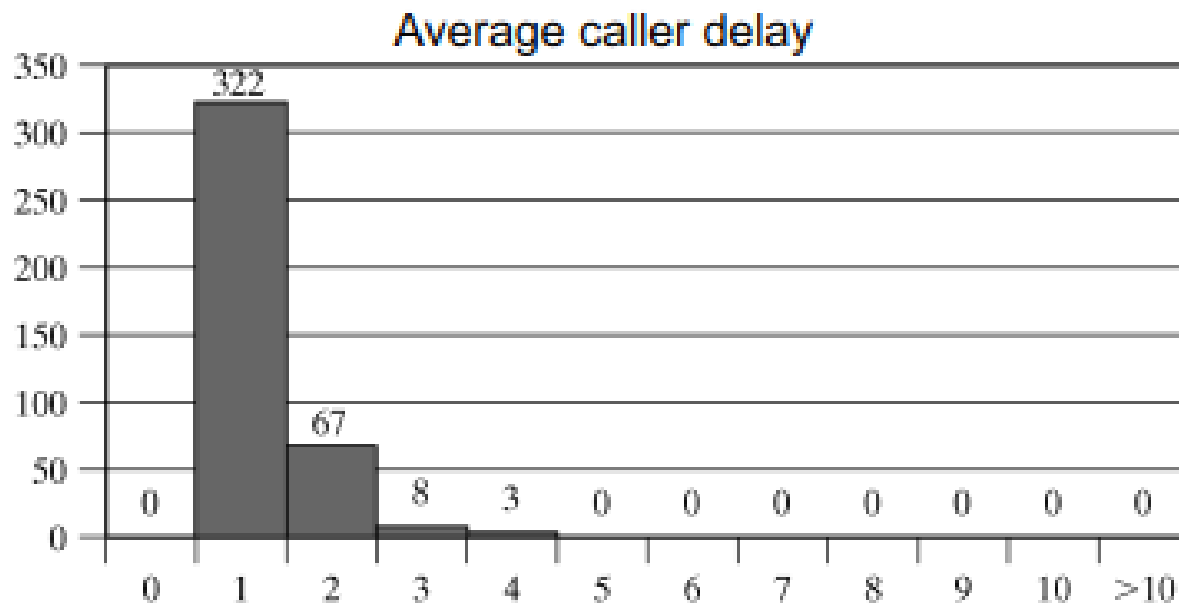


400 SIMULATION TRIALS OF 100 CALLER:

1. Average waiting time = 2 minutes.
2. Average waiting time = 1.7 minutes.
3. Average waiting time = 1.9 minutes.
4. Average waiting time = 2.5 minutes.
5. Average waiting time = 1.5 minutes.
- ...
400. Average waiting time = 2.7 minutes.

400 SIMULATION TRIALS OF 100 CALLER:

- 80.5% of callers had delay up to 1 minute.
- 19.5% of callers had delay more than 1 minute.





Thanks
Any questions